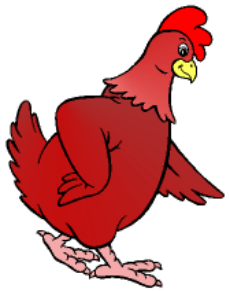
sihT esraP oT yrT

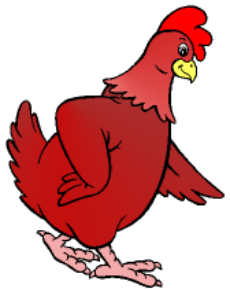
The Cost of Spookiness

$$|\,\rangle\!\rangle - |\,\rangle\!\rangle$$





Dave Bacon
Santa Fe Institute



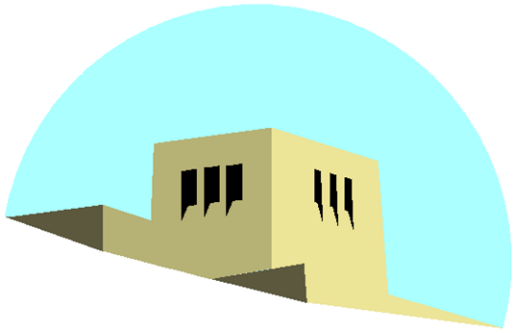
SFI Research Talk, New Years Day, 4703

In Collaboration With…

G.S. Ben Toner– Caltech
G.S. Tracey Tessier – Univ. of New Mexico
Prof. Carl Caves – Univ. of New Mexico
Prof. Ivan Deutsch – Univ. of New Mexico
G.S. Brian Eastin – Univ. of New Mexico

CALTECH





The University of New Mexico

Propaganda



quantum



Niels Bohr

They understood it.

They used it.

n safely say that nobody
ds quantum mechanics.

They just needed to ditch their preconceptions.

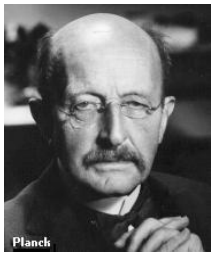
War Is Over

Inventing

Worrying

Accepting

190 191 192 193 194 195 196 197 198 199 200



Max Planck



Niels Bohr



Louis de Broglie



Erwin Schrodinger



Werner Heisenberg



John Bell



David Bohm



Peter Shor



Charlie Bennett

Worrying?



"I have this student, he's thinking about the foundations of quantum mechanics. He is doomed." — **John McCarthy** (of Artificial Intelligence fame)

Quantum Information Science

…is what happens when we stop worrying about quantum theory and love the strangeness



Were the worries were in vain?

This talk: **NO**

Talk To Me



ALICE

a

Feq(a,b) =

1 if a=b
0 if a≠b



BOB

b

0	1	0	1	•••	0

n bits

0	1	1	1	•••	0

n bits

How much (many bits) do Alice and Bob need
to exchange to compute F(a,b)?

With no error: n bits

With one-sided error ε: O(log(n/ε)) bits

Talk To Me Some More



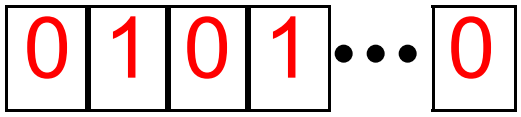
ALICE

a

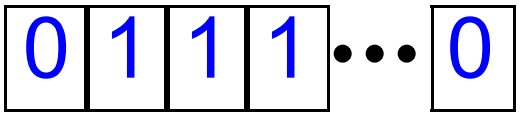


BOB

b



n bits



n bits

Ixor(a,b)=a1⊕a2 ⊕ …⊕an⊕b1⊕b2 ⊕ …⊕bn

How much (many bits) do Alice and Bob need to exchange to compute F(a,b)?

With no error.

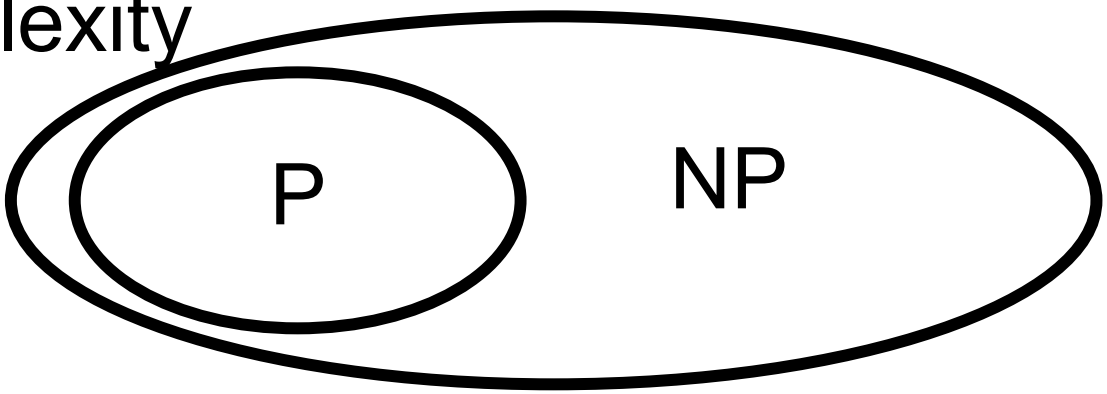
Communication Complexity

How much communication between Alice and Bob is necessary for them to compute F(a,b) when Alice knows a and Bob knows b?

[Yao 79]

many rigorous lower and matching upper bounds

…in contrast to computational complexity



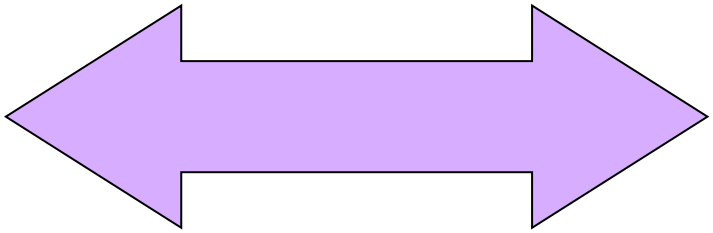
N=1
?
P=0?

Make It Quantum



ALICE

a



communicate qubits

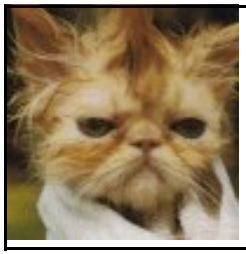
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



BOB

b

How does classical communication complexity (bits) compare to quantum communication complexity (qubits)?



Holevo's Theoerem: sending n classical bits requires n quantum bits.

Not So Fast



Ran Raz
(1999)

There exists functions F(a,b) which require Ω(m1/4/log(m)) bits of communication to solve
but can be solved using only O(log(m)) qubits.
[m is the length of a and b]



Exponential benefit in quantum over classical.

Exact model 0-error:
Buhrman, Cleve, Wigderson (1998)
1-way communication exponential separation:
Bar-Yossef, Jayram, Kerenidis (2004)

Quantum Teleportation



ALICE

a



BOB

b

1

qubits

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$\cong$



ALICE

a



BOB

b

2

bits

entangled quantum state

$|\psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

Quantum Teleportation Protocol

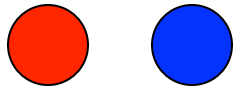
Entanglement as a Resource

If parties share entangled states, they can outperform classical communication complexity protocols. What is entanglement?



ALICE

Alice performs measurement of her choice.



Bob performs measurement of his choice.



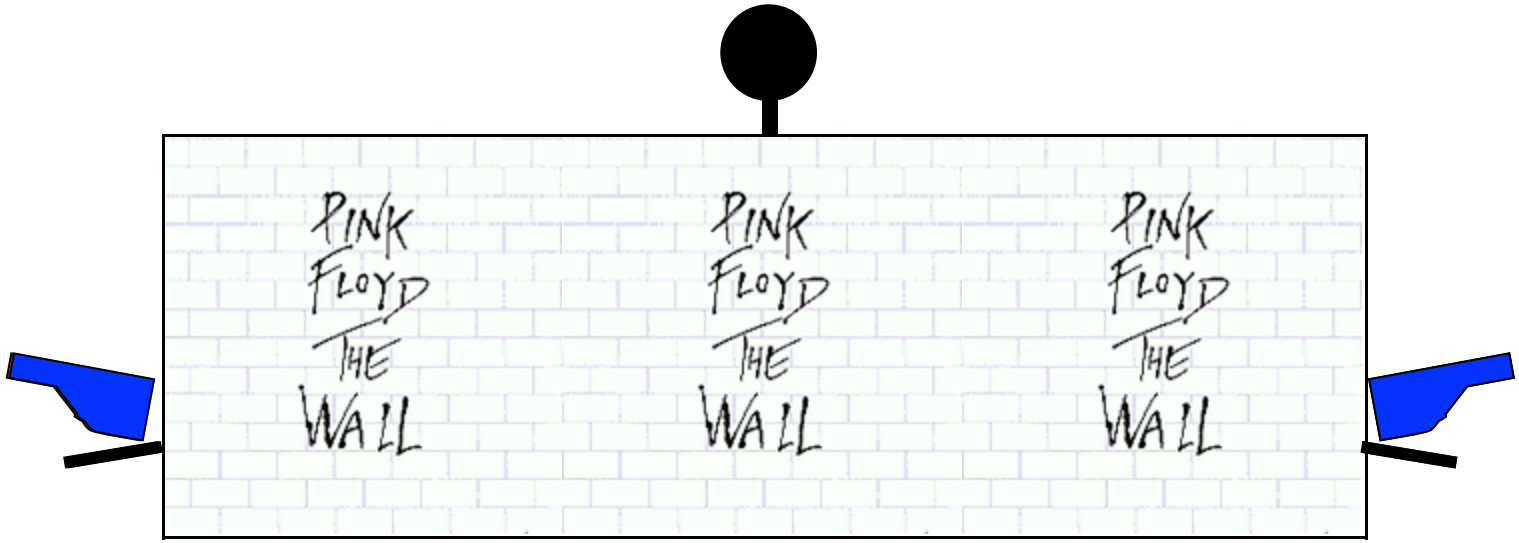
BOB

Alice gets measurement outcome

CORRELATED

Bob gets measurement outcome

What Entanglement Is Not



shared
randomness
once shoes put on, cannot change
(shoes not allowed to talk to each other)

note: shared randomness cannot be used to communicate

For Whom the Bell Tolls



A happy
John Bell

Entanglement is **not** like shared randomness

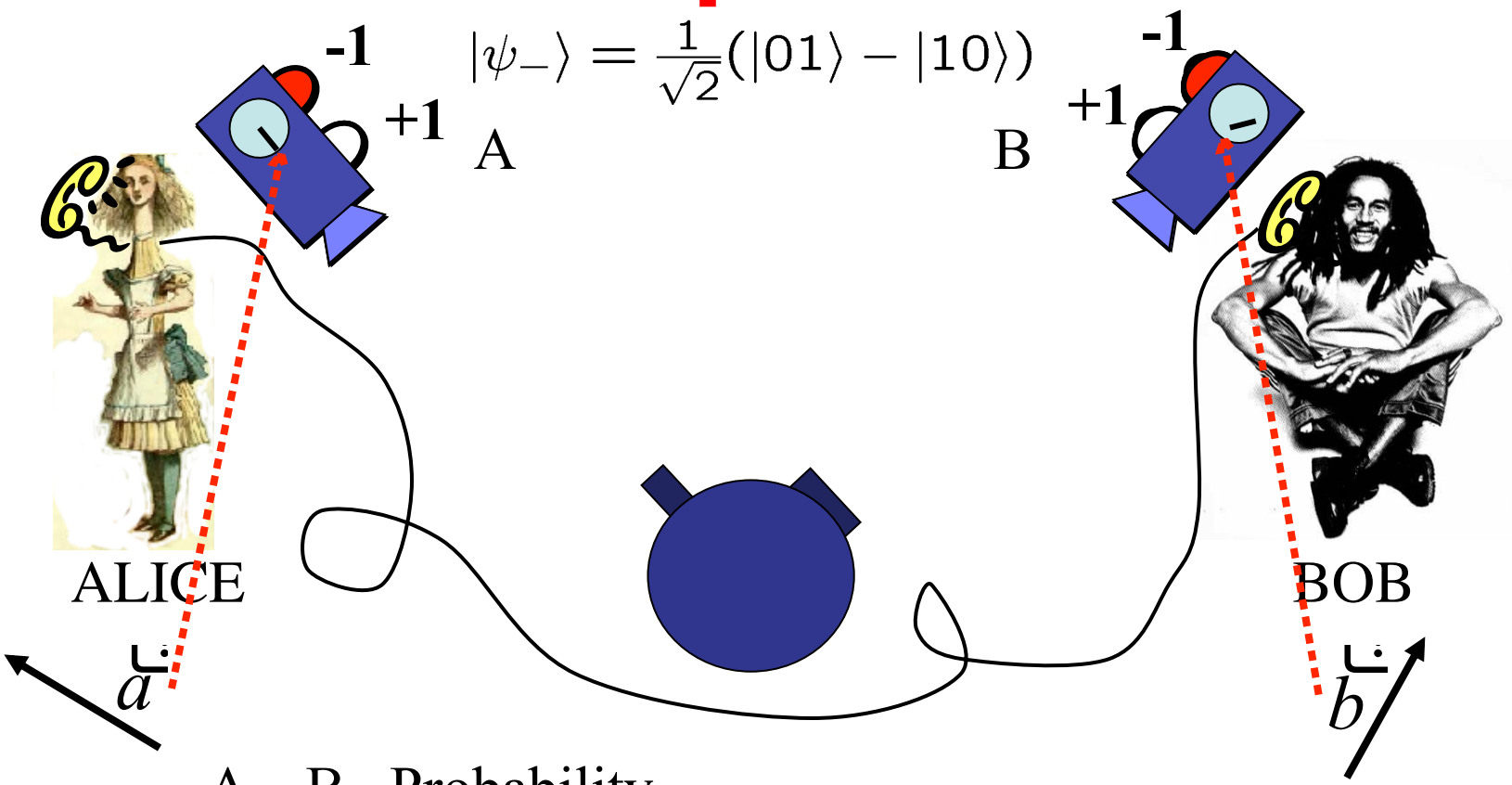
Historical context:

Einstein: quantum theory is incomplete
there is a deeper theory which gives rise to quantum theory

$|\psi\rangle \Longleftrightarrow$ realistic variables of a more fundamental theory.

John Bell to Einstein: “If those variables are local, then such a
fundamental theory cannot exist.”

Bell Experiment



A	B	Probability
- 1	- 1	$\frac{1}{4}\left(1-\vec{a}\cdot\vec{b}\right)$
1	+1	$\frac{1}{4}\left(1+\vec{a}\cdot\vec{b}\right)$
+1	- 1	$\frac{1}{4}\left(1+\vec{a}\cdot\vec{b}\right)$
+1	+1	$\frac{1}{4}\left(1-\vec{a}\cdot\vec{b}\right)$

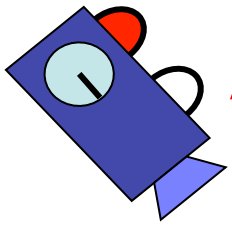
$$\langle A\rangle = \langle B\rangle = 0$$

$$\langle AB\rangle = -\vec{a}\cdot\vec{b}$$

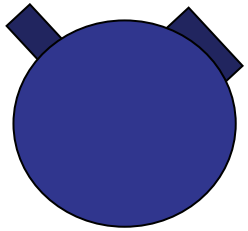
Bell Inequality (CHSH)



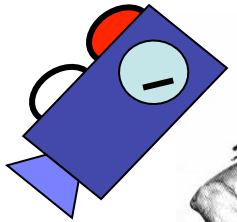
ALICE



$A \in \{+1,-1\}$



$B \in \{+1,-1\}$





BOB

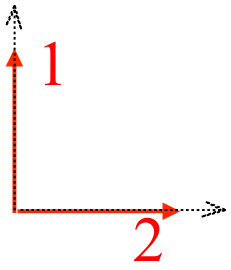
$$|A1B1+A1B2+A2B1-A2B2| \leq 2$$

Consider all deterministic values for A1,A2,B1,B2 where parties output is independent of others measurement

$$A1(B1+B2)+A2(B1-B2)$$

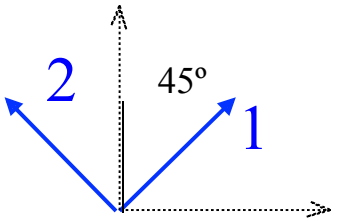
Probabilistic= Convex sum of deterministic

Celebrate Good Times



$$\langle AB \rangle = -\vec{a} \cdot \vec{b}$$

	1	2
1	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$
2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$



Quantum:

$$|A1B1+A1B2+A2B1-A2B2| = 2\sqrt{2}$$

Local Hidden Variables:

$$|A1B1+A1B2+A2B1-A2B2| \leq 2$$

Bell's Theorem: Quantum theory cannot be explained by a local hidden variable theory.

Recap

1. Entanglement can be used for exponential gain in quantum communication complexity.

2. Entanglement is not like shared randomness.

3. Physicists have worried about entanglement since it invalidates local realism. (spooky correlation)

Quantum information science: I don't care about locality, I just care about these using these strange quantum correlations as a resource!



The Question

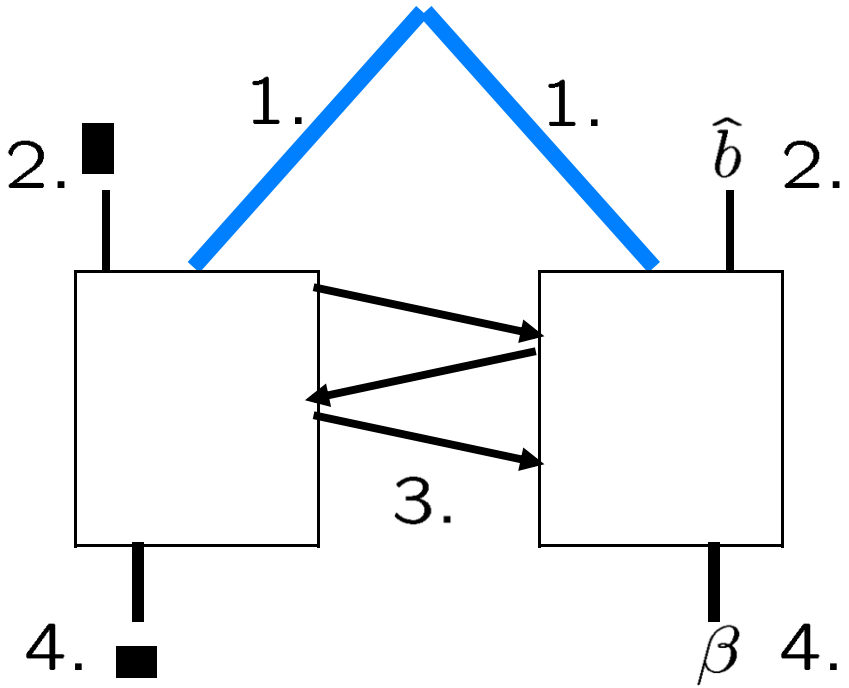
What is the communication cost of simulating the correlations produced by measuring entangled quantum states?

What are the COSTS of producing quantum correlations?



Quantify Bell's Theorem

Simulating Bell Correlations



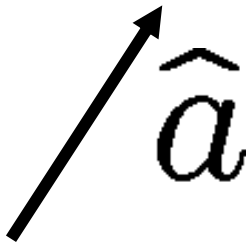
1. distribute shared random bits
2. A and B choose measurement directions, $\hat{a}$, $\hat{b}$
3. A and B communicate b bits
4. A and B output $\alpha, \beta \in \{\pm 1\}$

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta\hat{a}\cdot\hat{b}\right)$$

Why Is It Hard?

$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

How to produce distribution without sending $\hat{a}$ from A to B (or vice versa)?



requires infinite number of bits to specify

Towards Simulating Bell

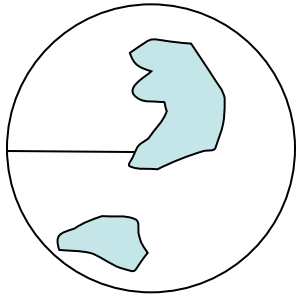
$$\Pr(\alpha, \beta | \hat{a}, \hat{b}) = \frac{1}{4}\left(1 - \alpha\beta \hat{a} \cdot \hat{b}\right)$$

Notice:

1. If $\hat{a} = \hat{b}$, then $Pr(\alpha = \beta) = 0$

   If $\hat{a} = -\hat{b}$, then $Pr(\alpha = \beta) = 1$

If B learns from A that A's vector lies in some region



and B knows A's output in this region

$\Downarrow$

forces B's output

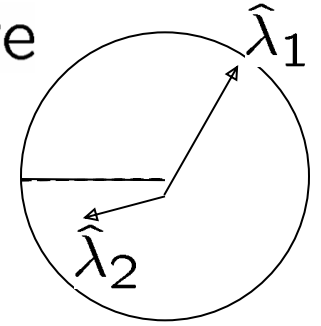
2. $\Pr\left(\alpha, \beta | \hat{a}, \hat{b}\right)$ is symmetric under joint rotations.

$$\Pr\left(\alpha, \beta | \hat{a}, \hat{b}\right) = \Pr\left(\alpha, \beta | R(\hat{a}), R(\hat{b})\right)$$

A 1 Bit Protocol

Two shared random variables:

$\hat{\lambda}_1, \hat{\lambda}_2$ uniformly distributed on sphere



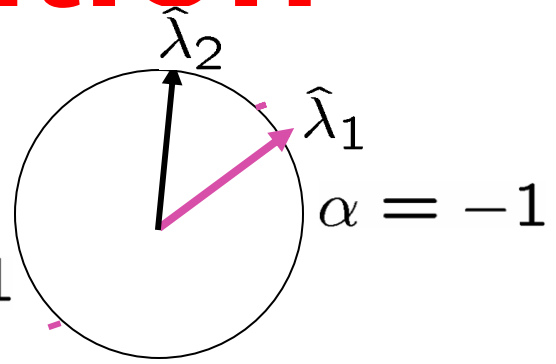
1. Alice outputs $-\text{sgn}\left(\hat{a} \cdot \hat{\lambda_2}\right)$

2. Alice sends a bit to Bob: $c = \text{sgn}\left(\hat{a} \cdot \hat{\lambda}_1\right) \text{sgn}\left(\hat{a} \cdot \lambda_2\right)$

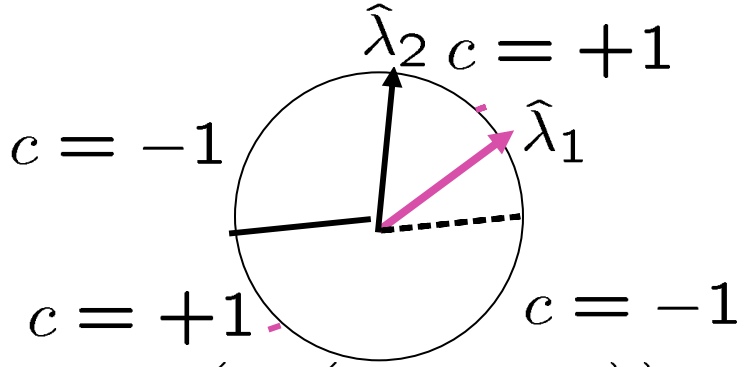
3. Bob outputs $\text{sgn}\left(\hat{b} \cdot \left(\hat{\lambda_1} + c\hat{\lambda_2}\right)\right)$

The Intuition

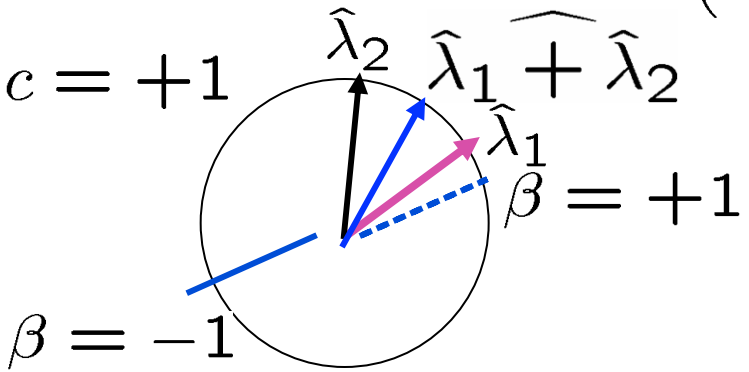
1. Alice outputs $-\text{sgn}\left(\hat{a}\cdot\hat{\lambda_1}\right)$

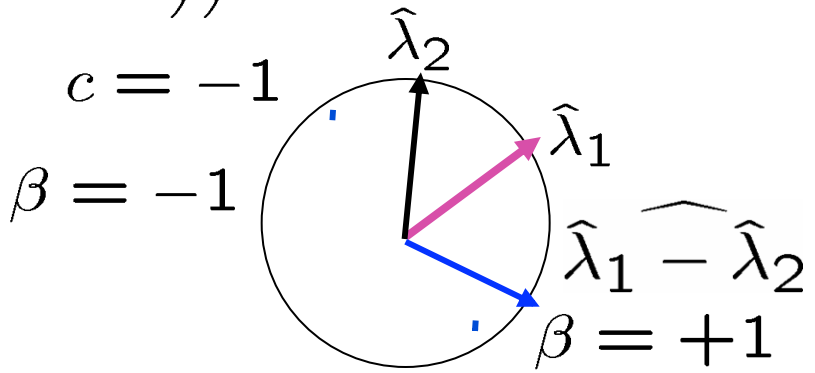


2. Alice sends a bit to Bob: $c=\text{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\text{sgn}\left(\hat{a}\cdot\lambda_2\right)$



3. Bob outputs $\text{sgn}\left(\hat{b}\cdot\left(\hat{\lambda_1}+c\hat{\lambda_2}\right)\right)$





It Works!

Each parties output changes sign under

$$\hat{\lambda}_1 \to -\hat{\lambda}_1,\ \hat{\lambda}_2 \to -\hat{\lambda}_2 \quad \Rightarrow \langle\alpha\rangle = \langle\beta\rangle = 0$$

$$\langle\alpha\beta\rangle = E\left[-\mathrm{sgn}(\hat{a}\cdot\hat{\lambda}_1)\sum_{d=\pm 1}\frac{1+cd}{2}\,\mathrm{sgn}\left[\hat{b}\cdot(\hat{\lambda}_1+d\hat{\lambda}_2)\right]\right]$$

$$E[x] = \frac{1}{(4\pi)^2}\int d\hat{\lambda}_1\int d\hat{\lambda}_2[x] \qquad c = \mathrm{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\mathrm{sgn}\left(\hat{a}\cdot\lambda_2\right)$$

$$\langle\alpha\beta\rangle = E\left[-2\,\mathrm{sgn}(\hat{a}\cdot\hat{\lambda}_1)\,\mathrm{sgn}\left[\hat{b}\cdot(\hat{\lambda}_1+\hat{\lambda}_2)\right]\right]$$

$$\frac{1}{4\pi}\int d\hat{\lambda}_2 \mathrm{sgn}\left(r+\hat{b}\cdot\hat{\lambda}_2\right) = r$$

$$-\frac{1}{4\pi}\int d\hat{\lambda}_1 2\mathrm{sgn}\left(\hat{a}\cdot\hat{\lambda}_1\right)\hat{b}\cdot\hat{\lambda}_1 = -\hat{a}\cdot\hat{b}$$

$$\langle\alpha\beta\rangle = -\hat{a}\cdot\hat{b}$$

1 Bell Pair's Worth of Correlations

correlations from projective measurements on Bell pair

$=$

shared randomness

$+$

1 bit of communication

B. Toner and D. Bacon, *Phys. Rev. Let.,* **91**, 187904 (2003)

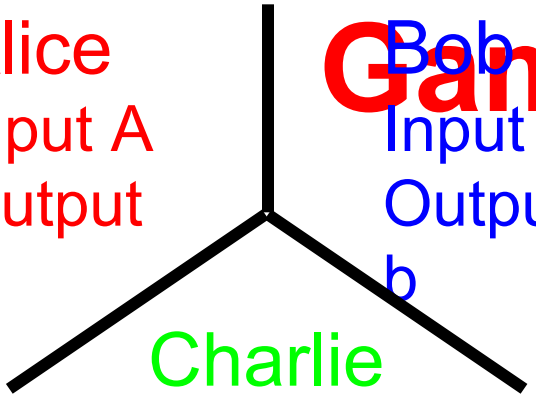
The Finiteness Conjecture

Conjecture: all multipartite entangled systems can be exactly simulated by augmenting a local hidden variable theory with an amount of communication polynomial in the product of the dimensions of the entangled subsystems.

- projective measurements on two entangled qubits (Brassard, Cleve, Tapp 99, Toner, Bacon 03)
- all bipartite two outcome correlation observables (Bacon, Toner, t.b.p., 05)

Today: add more evidence for the conjecture.

GHZ/Mermin Game



Input: X or Y, Output: +1 or -1

Win condition:

A	B	C	abc
X	X	X	+1
Y	Y	X	-1
Y	X	Y	-1
X	Y	Y	-1

Classical

aY bY cX = -1

aY bX cY = -1

aX bY cY = -1

× ——————

aX bX cX = -1

➡ not possible to always win

Quantum



$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

XXX|GHZ⟩ =|GHZ⟩

YYX|GHZ⟩ =-|GHZ⟩

➡ possible to always win

Cheaters Always Win



Alice
Input A
Output
a

Bob
Input B
Output
b



Charlie
Input C
Output c

Win condition:

A	B	C	abc
X	X	X	+1
Y	Y	X	-1
Y	X	Y	-1
X	Y	Y	-1

Charlie: always outputs c=+1

Bob: always outputs b=+1, tells Alice his measurement

Alice: knows A and B

A	B	a	abc
X	X	+1	+1
Y	Y	-1	-1
Y	X	-1	-1
X	Y	-1	-1

Using a single bit of communication they can win.

GHZ Extended

Alice	Bob
Input A	Input B
Output a	Output b
Charlie	**Dodo**
Input C	Input D
Output c	Output d

win condition

A	B	C	D	abcd
X	X	X	X	+1
X	X	Y	Y	-1
X	Y	Y	X	-1
X	Y	X	Y	-1

Quantum

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$$

possible to always win

Classical (no talking)

aX bX cY dY= -1

aX bY cX dY= -1

aX bY cY dX= -1

× ——————

aX bX cX dX= -1

impossible to always win

Classical (A and B talk)

vXXcY dY= -1

v̄XYcY dX= -1

vXYcX dY= -1

× ——————

vXXcX dX= -1

impossible to always win

GHZ Super Extended

win condition

A	B	C	D	abcd
X	X	X	X	+1
X	X	Y	Y	-1
X	Y	Y	X	-1
X	Y	X	Y	-1
Y	X	X	Y	-1
Y	X	Y	X	-1
Y	Y	X	X	-1

Quantum

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$$

possible to always win

Alice	Bob
Input A	Input B
Output a	Output b
Charlie	**Dodo**
Input C	Input D
Output c	Output d

If any two parties talk, they cannot always win!

Bell Inequality With Communication

two parties, three two-outcome measurements
local hidden variable theory + 1 bit of communication

D. Bacon and B.F. Toner, *Phys. Rev. Let.* **90**, 157904 (2003)

$$|-ab+ac-ba+bb+bc+ca+cb+cc| \leq 6$$

$$|aa+2ab-2ac+2ba+bb+2bc-2ca+2cb+cc| \leq 11$$



but, quantum theory has

$$\langle -AB+AC-BA+BB+BC+CA+CB+CC \rangle \quad QM \leq 6$$

$$\langle AA+2AB-2AC+2BA+BB+2BC-2CA+2CB+CC \rangle$$

$$QM \leq 11$$

four parties, two two-outcome measurements
local hidden variable theory + 1 bit of communication

$$|aaaa-aabb-abab-abba-baab-baba-bbaa| \leq 5$$



quantum theory violates this bound

$$\langle AAAA-AABB-ABAB-ABBA-BAAB-BABA-BBAA \rangle$$

Measure It!

Just as Bell Inequalities tell us that we can experimentally distinguish between quantum theory and local hidden variable models,

this Bell Inequality with Communication tells us that we can experimentally distinguish between quantum theory and local hidden variable models, plus one bit of cheating communication.

$$|aaaa-aabb-abab-abba-baab-baba-bbaa| \leq 5$$

$$\langle AAAA-AABB-ABAB-ABBA-BAAB-BABA-BBAA \rangle_{GHZ} = 7$$

3,4,…,n

win condition

A …….	n-3	n-2	n-1	n	abcd
X …….	X	X	X	X	+1
X …….	X	X	Y	Y	-1
X …….	X	Y	Y	X	-1
X …….	X	Y	X	Y	-1

local hidden variable theory
plus n-3 bits of communication

$|A\ldots A-\sum(\text{observables with 2 B's and rest A's})|\leq n(n-1)/2-1$

$\langle A\ldots A-\sum(\text{observables with 2 B's and rest A's})\rangle$ GHZ=$n(n-1)/2+1$

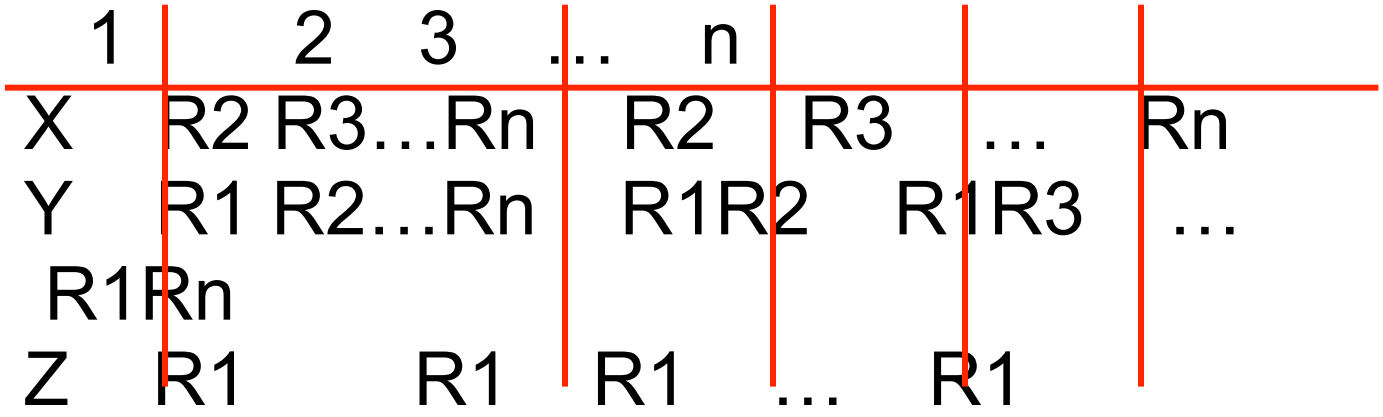
Bell Inequality with n-3 bits of communication

Simulating Pauli Measurements on GHZ

n parties, each measure {X,Y,Z} on n party GHZ state

BIwC implies least n-2 bits of communication necessary

n-2 bits of communication is also sufficient



$R_i \in \{+1,-1\}$

uniform random variable s

parties 2…n-1 communicate to party 1 whether they measured Y or not

party 1 flips her outcome bit if [hers Y plus 2…n-1 Y's] is odd else she does nothing

The Finiteness Conjecture

Conjecture: all multipartite entangled systems can be exactly simulated by augmenting a local hidden variable theory with an amount of communication polynomial in the product of the dimensions of the entangled subsystems.

- projective measurements on two entangled qubits (Brassard, Cleve, Tapp 99, Toner, Bacon 03)
- all bipartite two outcome correlation observables (Bacon, Toner 05)
- Pauli measurements on a n-party GHZ states (Tessier, Caves, Deutsch, Eastin, Bacon 05)

Conclusion

- Entanglement gives exponential benefits
- Entanglement is not like shared classical randomness

- What is the communication cost of simulating quantum correlations?

- Simulating projective measurement on a singlet requires shared randomness plus a single bit of communication

- Simulating Pauli measurement on a n-party GHZ state requires n-2 bits of communication and this is also sufficient.

- Is there a deeper structure to quantum correlations? Can all correlations be simulated by shared randomness and an amount of communication related to the amount of entanglement?